

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456738.748708 +/- 3.5e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.045
Transit depth (Rp/Rs)^2	2.38 +/- 1.4 [%]
Orbital Inclination (inc)	83.26 +/- 2.91
Airmass coefficient 1 (a1)	1.71 +/- 0.5
Airmass coefficient 2 (a2)	-0.203 +/- 0.037
Scatter in the residuals of the lightcurve fit is	24.93 %
Best Comparison Star	None
Optimal Aperture	1.94
Optimal Annulus	7.95
Transit Duration (day)	0.1282 +/- 0.0087

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.045 vs 0.1178 (deviation 0.48σ)
Duration fit vs published	0.1282 d vs 0.125 d (deviation 0.22σ)
Mid-time O–C	0.1 min
Depth SNR	2.0
Residual scatter	24.93 %

Light curve

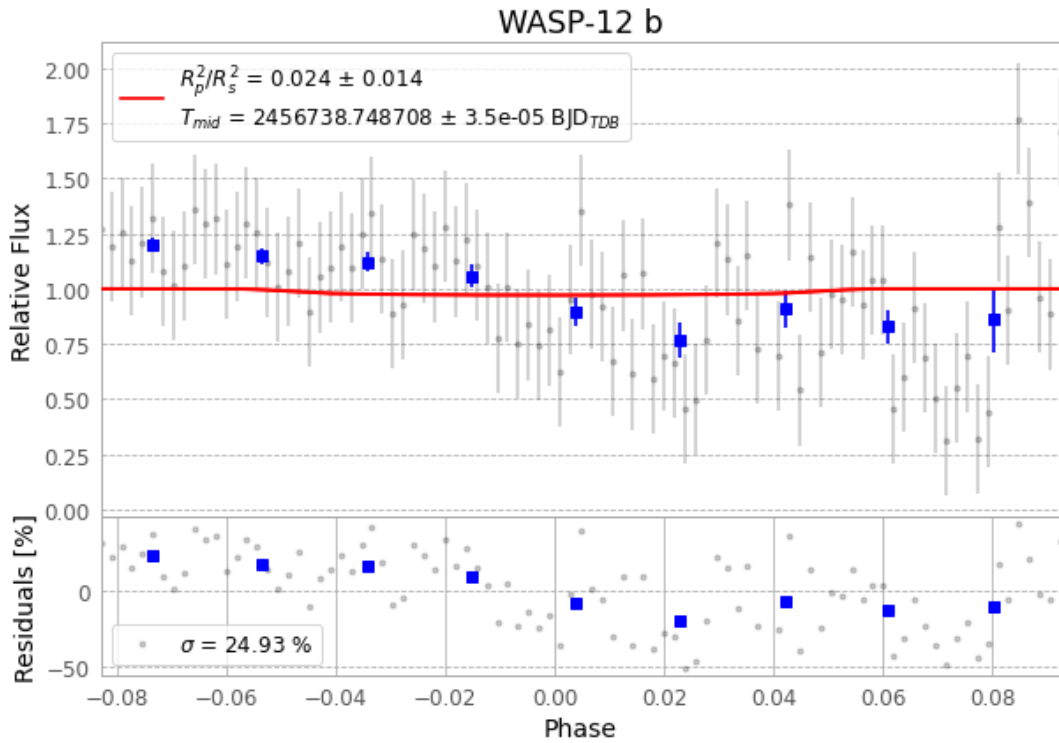


Figure 1 — FinalLightCurve_WASP-12 b_2014-03-21.png (SHA-256 0f05bda7498214bd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [356, 292], 1 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 9a2d2e2a5a88bf6d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

93 FITS frames (61 MB), WASP-12140322034513.FITS → WASP-12140322082112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-140322.log (487651bc99af...), FinalLightCurve_WASP-12 b_2014-03-21.png (0f05bda74982...), FinalLightCurve_WASP-12 b_2014-03-21.csv (135c2865332c...)

Compute resources

Wall time 92.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 23:17Z.